

Exercise 1.5Question 1 (i)

Solve the following system by Gaussian elimination method:

$$2x - 2y + 3z = 2$$

$$x + 2y - z = 3$$

$$3x - y + 2z = 1$$

Solution:

Augmented matrix

$$[A|B] = \left[\begin{array}{ccc|c} 2 & -2 & 3 & 2 \\ 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \end{array} \right]$$

Interchange R_1 and R_2

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -2 & 3 & 2 \\ 3 & -1 & 2 & 1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & -7 & 5 & -8 \end{array} \right]$$

$$R_3 \rightarrow 6R_3 - 7R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & 0 & -5 & -20 \end{array} \right]$$

$$-5z = -20, \text{ so } z = 4$$

$$-6y + 5z = -4 \Rightarrow -6y + 20 = -4 \Rightarrow y = 4$$

$$x + 2y - z = 3 \Rightarrow x + 8 - 4 = 3 \Rightarrow x = -1$$

Therefore, $x = -1$, $y = 4$, $z = 4$.

Exercise 1.5Question 1 (ii)

Solve the following system by Gaussian elimination method:

$$2x + 4y + 6z = 22$$

$$3x + 8y + 5z = 27$$

$$-x + y + 2z = 2$$

Solution:

Augmented matrix

$$[A|B] = \left[\begin{array}{ccc|c} 2 & 4 & 6 & 22 \\ 3 & 8 & 5 & 27 \\ -1 & 1 & 2 & 2 \end{array} \right]$$

$$R_1 \rightarrow \frac{R_1}{2} \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 3 & 8 & 5 & 27 \\ -1 & 1 & 2 & 2 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 3R_1, \quad R_3 \rightarrow R_3 + R_1 \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 2 & -4 & -6 \\ 0 & 3 & 5 & 13 \end{array} \right]$$

$$R_2 \rightarrow \frac{R_2}{2} \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & -2 & -3 \\ 0 & 3 & 5 & 13 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 3R_2 \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 11 & 22 \end{array} \right]$$

$$11z = 22, \text{ so } z = 2$$

$$y - 2z = -3 \Rightarrow y - 4 = -3 \Rightarrow y = 1$$

$$x + 2y + 3z = 11 \Rightarrow x + 2 + 6 = 11 \Rightarrow x = 3$$

Therefore, $x = 3$, $y = 1$, $z = 2$.

Exercise 1.5Question 2

If $ax^2 + bx + c$ is divided by $x+3$, $x-5$, and $x-1$, the remainders are 21, 61 and 9 respectively. Find a , b and c using Gaussian elimination method.

Solution:

Let $f(x) = ax^2 + bx + c$

By remainder theorem:

$$f(-3) = 21 \Rightarrow 9a - 3b + c = 21$$

$$f(5) = 61 \Rightarrow 25a + 5b + c = 61$$

$$f(1) = 9 \Rightarrow a + b + c = 9$$

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 9 & -3 & 1 & 21 \\ 25 & 5 & 1 & 61 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 - 9R_1 \\ R_3 \rightarrow R_3 - 25R_1 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -12 & -8 & -60 \\ 0 & -20 & -24 & -164 \end{array} \right]$$

Exercise 1.5Question 2 - continued

$$\begin{array}{l} R_2 \rightarrow -R_2/4 \\ R_3 \rightarrow -R_3/4 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & 3 & 2 & 15 \\ 0 & 5 & 6 & 41 \end{array} \right]$$

$$R_3 \rightarrow 3R_3 - 5R_2 \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & 3 & 2 & 15 \\ 0 & 0 & 8 & 48 \end{array} \right]$$

$$8c = 48, \text{ so } c = 6$$

$$3b + 2c = 15 \Rightarrow 3b + 12 = 15 \Rightarrow b = 1$$

$$a + b + c = 9 \Rightarrow a + 1 + 6 = 9 \Rightarrow a = 2$$

Therefore, $a = 2$, $b = 1$, $c = 6$.

Exercise 1.5Question 3

An amount of Rs 65,000 is invested in three bonds at the rates of 6%, 8% and 10% per annum respectively. The total annual income is Rs 5,000. The income from the third bond is Rs 800 more than that from the second bond. Determine the amount invested in each bond using Gaussian elimination method.

Solution:

Let the amounts invested be x , y and z .

$$x + y + z = 65000$$

$$\frac{6x}{100} + \frac{8y}{100} + \frac{10z}{100} = 5000$$

$$\frac{10z}{100} = 800 + \frac{8y}{100}$$

$$\text{So, } 6x + 8y + 10z = 500000$$

$$\text{and } -8y + 10z = 80000$$

Augmented matrix

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 6 & 8 & 10 & 500000 \\ 0 & -8 & 10 & 80000 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 6R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 0 & 2 & 4 & 110000 \\ 0 & -8 & 10 & 80000 \end{array} \right]$$

Exercise 1.5Question 3 - continued

From the previous step:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 0 & 2 & 4 & 110000 \\ 0 & -8 & 10 & 80000 \end{array} \right]$$

$$R_3 \rightarrow R_3 + 4R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 0 & 2 & 4 & 110000 \\ 0 & 0 & 26 & 520000 \end{array} \right]$$

$$26z = 520000, \text{ so } z = 20000$$

$$\text{From second row: } 2y + 4z = 110000$$

$$2y + 80000 = 110000$$

$$2y = 30000, \text{ so } y = 15000$$

$$\text{From first row: } x + y + z = 65000$$

$$x + 15000 + 20000 = 65000$$

$$x = 30000$$

Therefore, $x = 30000$, $y = 15000$, $z = 20000$.

Exercise 1.5Question 4

A boy is walking along the path $y = ax^2 + bx + c$ through the points $(-6, 8)$, $(-2, -12)$, and $(3, 8)$. He wants to meet his friend at $P(7, 60)$. Will he meet his friend? Use Gaussian elimination method.

Solution:

$$y = ax^2 + bx + c$$

$$\text{At } (-6, 8): \quad 36a - 6b + c = 8$$

$$\text{At } (-2, -12): \quad 4a - 2b + c = -12$$

$$\text{At } (3, 8): \quad 9a + 3b + c = 8$$

Augmented matrix

$$[A|B] = \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 4 & -2 & 1 & -12 \\ 9 & 3 & 1 & 8 \end{array} \right]$$

Exercise 1.5Question 4 - continued

From the augmented matrix

$$\left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 4 & -2 & 1 & -12 \\ 9 & 3 & 1 & 8 \end{array} \right]$$

$$R_2 \rightarrow R_1 - 9R_2$$

$$R_3 \rightarrow 4R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & 12 & -8 & 116 \\ 0 & 18 & 3 & 24 \end{array} \right]$$

$$R_2 \rightarrow R_2/4$$

$$R_3 \rightarrow R_3/3$$

$$\left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & 3 & -2 & 29 \\ 0 & 6 & 1 & 8 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & 3 & -2 & 29 \\ 0 & 0 & 5 & -50 \end{array} \right]$$

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Exercise 1.5

Question 4 - final answer

From the third row: $5c = -50$, so $c = -10$

From the second row: $3b - 2c = 29$

$$3b - 2(-10) = 29$$

$$3b + 20 = 29$$

$$3b = 9, \text{ so } b = 3$$

From the first row: $36a - 6b + c = 8$

$$36a - 18 - 10 = 8$$

$$36a = 36, \text{ so } a = 1$$

Therefore the path is $y = x^2 + 3x - 10$

At $P(7, 60)$:

$$y = 1(7^2) + 3(7) - 10$$

$$y = 49 + 21 - 10 = 60$$

Hence, he will meet his friend at $P(7, 60)$.

Exercise 1.5 - Final Answers

Question 1 (i): $x = -1, y = 4, z = 4$

Question 1 (ii): $x = 3, y = 1, z = 2$

Question 2: $a = 2, b = 1, c = 6$

Question 3: $x = 30000, y = 15000, z = 20000$

Question 4: $a = 1, b = 3, c = -10$

So the path is $y = x^2 + 3x - 10$

At $P(7, 60), y = 60$

Therefore, the boy will meet his friend at $P(7, 60)$.

Completed step by step in 10 handwritten pages.

Exercise 1.6

Question 1

Test for consistency and if possible, solve the following systems of equations by rank method.

$$(i) \quad x - y + 2z = 2, \quad 2x + y + 4z = 7, \quad 4x - y + z = 4$$

Solution:

Matrix form

$$[A] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 4 \end{bmatrix} \quad \text{where} \quad A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 4 \\ 4 & -1 & 1 \end{bmatrix}$$

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 2 & 1 & 4 & 7 \\ 4 & -1 & 1 & 4 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 4R_1$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 3 & 0 & 3 \\ 0 & 3 & -7 & -4 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 3 & 0 & 3 \\ 0 & 0 & -7 & -7 \end{array} \right]$$

Therefore, $\text{rank}(A) = 3$ and $\text{rank}([A|B]) = 3 = n$

The system is consistent and has a unique solution.

Writing the equivalent equations from echelon form:

$$x - y + 2z = 2 \quad \dots (1)$$

$$3y = 3 \Rightarrow y = 1$$

$$-7z = -7 \Rightarrow z = 1$$

$$\text{From (1): } x - 1 + 2 = 2 \Rightarrow x = 1$$

Hence, $x = 1, y = 1, z = 1$.

Exercise 1.6Question 1 (ii)

$$3x + y + z = 2, \quad x - 3y + 2z = 1, \quad 7x - y + 4z = 5$$

Solution:

Matrix form

$$AX = B \quad \text{where} \quad A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & -3 & 2 \\ 7 & -1 & 4 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$$

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} 3 & 1 & 1 & 2 \\ 1 & -3 & 2 & 1 \\ 7 & -1 & 4 & 5 \end{array} \right]$$

$$R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 3 & 1 & 1 & 2 \\ 7 & -1 & 4 & 5 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 3R_1, \quad R_3 \rightarrow R_3 - 7R_1$$

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 0 & 10 & -5 & -1 \\ 0 & 20 & -10 & -2 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 0 & 10 & -5 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Therefore, $\text{rank}(A) = 2$ and $\text{rank}([A | B]) = 2 < n$.

The system is consistent and has infinitely many solutions.

Writing the equivalent equations from echelon form:

$$x - 3y + 2z = 1 \quad \dots (1)$$

$$10y - 5z = -1 \quad \dots (2)$$

Put $z = t$. Continue on next page.

Exercise 1.6Question 1 (ii) – continued

From (2): $10y - 5z = -1$
 $10y = -1 + 5z = 5t - 1$

Therefore $y = \frac{5t - 1}{10}$.

From (1): $x - 3y + 2z = 1$
 $x - 3\left(\frac{5t - 1}{10}\right) + 2t = 1$
 $x = 1 + 3\left(\frac{5t - 1}{10}\right) - 2t$
 $x = \frac{7 - 5t}{10}$.

Hence, the parametric solution is

$$(x, y, z) = \left(\frac{7 - 5t}{10}, \frac{5t - 1}{10}, t \right), \text{ for all } t \in \mathbb{R}.$$

Question 1 (iii)

Solve the system of equations

$$2x + 2y + z = 5, \quad x - y + z = 1, \quad 3x + y + 2z = 4.$$

Solution:

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} 2 & 2 & 1 & 5 \\ 1 & -1 & 1 & 1 \\ 3 & 1 & 2 & 4 \end{array} \right]$$

$$R_1 \leftrightarrow R_2 \quad \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 2 & 2 & 1 & 5 \\ 3 & 1 & 2 & 4 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - 3R_1 \quad \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 4 & -1 & 3 \\ 0 & 4 & -1 & 1 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2 \quad \left[\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 4 & -1 & 3 \\ 0 & 0 & 0 & -2 \end{array} \right]$$

Exercise 1.6Question 1 (iii) — conclusion

Here $\text{rank}(A) = 2$ and $\text{rank}([A|B]) = 3$.

Since $\text{rank}(A)$ is not equal to $\text{rank}([A|B])$, the system is inconsistent.

Therefore, it has no solution.

Question 1 (iv)

$$2x - y + z = 2, \quad 6x - 3y + 3z = 6, \quad 4x - 2y + 2z = 4$$

Solution:

Matrix form

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 6 & -3 & 3 \\ 4 & -2 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$$

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 6 & -3 & 3 & 6 \\ 4 & -2 & 2 & 4 \end{array} \right]$$

$$R_2 \rightarrow R_2 / 3, \quad R_3 \rightarrow R_3 / 2$$

$$= \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 2 & -1 & 1 & 2 \\ 2 & -1 & 1 & 2 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$= \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Therefore, $\text{rank}(A) = 1$ and $\text{rank}([A|B]) = 1 < n$.

The system is consistent and has infinitely many solutions.

Continue the general solution on the next page.

Exercise 1.6Question 1 (iv) – continued

From the echelon form equation

$$2x - y + z = 2.$$

Put $y = s$ and $z = t$.

$$\text{So, } 2x - s + t = 2$$

$$2x = 2 + s - t$$

$$x = \frac{2 + s - t}{2}.$$

Therefore the general solution is

$$(x, y, z) = \left(\frac{2 + s - t}{2}, s, t \right), \text{ for all } s, t \in \mathbb{R}.$$

Summary for Question 1:

(i) Consistent with unique solution:

$$x = 1, y = 1, z = 1.$$

(ii) Consistent with infinitely many solutions:

$$(x, y, z) = \left(\frac{7 - 5t}{10}, \frac{5t - 1}{10}, t \right).$$

(iii) Inconsistent; no solution.

(iv) Consistent with infinitely many solutions:

$$(x, y, z) = \left(\frac{2 + s - t}{2}, s, t \right).$$

Exercise 1.6

Question 2

Find the value of k for which the equations

$$kx - 2y + z = 1,$$

$$x - 2ky + z = -2,$$

$$x - 2y + kz = 1$$

have (i) no solution, (ii) unique solution, (iii) infinitely many solutions.

Solution:

Matrix form

$$AX = B \quad \text{where} \quad A = \begin{bmatrix} k & -2 & 1 \\ 1 & -2k & 1 \\ 1 & -2 & k \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} k & -2 & 1 & 1 \\ 1 & -2k & 1 & -2 \\ 1 & -2 & k & 1 \end{array} \right]$$

$R_1 \leftrightarrow R_3$

$$\left[\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 1 & -2k & 1 & -2 \\ k & -2 & 1 & 1 \end{array} \right]$$

$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - kR_1$

$$\left[\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 0 & -2k+2 & 1-k & -3 \\ 0 & -2+2k & 1-k^2 & 1-k \end{array} \right]$$

$R_3 \rightarrow R_3 + R_2$

$$\left[\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 0 & 2(1-k) & 1-k & -3 \\ 0 & 0 & 2-k-k^2 & k+2 \end{array} \right]$$

$$2 - k - k^2 = (k+2)(1-k).$$

Case (i): If $k = 1$,

third row becomes $[0 \ 0 \ 0 \ | \ 3]$.

So $\text{rank}(A) = 2$, $\text{rank}([A|B]) = 3$.

Hence the system is inconsistent and has no solution.

Exercise 1.6

Question 2 — continued

Case (ii): If k is not equal to 1 and k is not equal to -2 ,
then $(k+2)(1-k)$ is not zero.

Therefore the third row is non-zero.

Hence $\text{rank}(A) = \text{rank}([A|B]) = 3 = n$.

So the system is consistent and has a unique solution.

Case (iii): If $k = -2$,

then the third row becomes $\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$,

while the second row remains non-zero.

Hence $\text{rank}(A) = \text{rank}([A|B]) = 2 < 3$.

So the system is consistent and has infinitely many solutions.

Final summary:

No solution when $k = 1$.

Unique solution when k is not equal to 1 and not equal to -2 .

Infinitely many solutions when $k = -2$.

Exercise 1.6Question 3

Investigate the values of λ and μ for which the system

$$2x + 3y + 5z = 9,$$

$$7x + 3y - 5z = 8,$$

$$2x + 3y + \lambda z = \mu$$

has (i) no solution, (ii) a unique solution, (iii) infinitely many solutions.

Solution:

Matrix form

We can write the system as $AX = B$, where

$$A = \begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -5 \\ 2 & 3 & \lambda \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 9 \\ 8 \\ \mu \end{bmatrix}$$

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 7 & 3 & -5 & 8 \\ 2 & 3 & \lambda & \mu \end{array} \right]$$

$$R_2 \rightarrow 2R_2 - 7R_1$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 0 & -15 & -45 & -47 \\ 2 & 3 & \lambda & \mu \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 0 & -15 & -45 & -47 \\ 0 & 0 & \lambda - 5 & \mu - 9 \end{array} \right]$$

So the echelon form is

$$\left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 0 & -15 & -45 & -47 \\ 0 & 0 & \lambda - 5 & \mu - 9 \end{array} \right]$$

Now discuss the three cases based on $\lambda - 5$ and $\mu - 9$.

Exercise 1.6Question 3 — cases (i) and (ii)

Case (i): No solution.

If $\lambda = 5$ and μ is not equal to 9,
then the third row becomes

$$[0 \quad 0 \quad 0 \mid \mu - 9],$$

which is impossible.

Hence $\text{rank}(A) = 2$ and $\text{rank}([A|B]) = 3$.

Therefore the system is inconsistent and has no solution.

Case (ii): Unique solution.

If λ is not equal to 5,
then the third row is non-zero.

Hence $\text{rank}(A) = \text{rank}([A|B]) = 3 = n$.

Therefore the system is consistent and has a unique solution.

In this case μ can be any real number.

Exercise 1.6

Question 3 — case (iii) and final summary

Case (iii): Infinitely many solutions.

If $\lambda = 5$ and $\mu = 9$,

then the third row becomes $[0 \ 0 \ 0 \mid 0]$.

Hence $\text{rank}(A) = \text{rank}([A|B]) = 2 < 3$.

Therefore the system is consistent and has infinitely many solutions.

Final summary for the whole exercise:

Question 1 (i): unique solution $x = 1, y = 1, z = 1$.

Question 1 (ii): infinitely many solutions with

$$(x, y, z) = \left(\frac{7-5t}{10}, \frac{5t-1}{10}, t \right).$$

Question 1 (iii): no solution.

Question 1 (iv): infinitely many solutions with

$$(x, y, z) = \left(\frac{2+s-t}{2}, s, t \right).$$

Question 2: no solution when $k = 1$;

unique solution when k is not equal to 1 and not equal to -2;

infinitely many solutions when $k = -2$.

Question 3: no solution when $\lambda = 5$ and μ is not equal to 9;

unique solution when λ is not equal to 5;

infinitely many solutions when $\lambda = 5$ and $\mu = 9$.

Completed in 10 handwritten pages.

Exercise 1.7

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Question 1

Solve the following system of homogeneous equations.

$$\begin{aligned} \text{(i)} \quad & 3x + 2y + 7z = 0, \\ & 4x - 3y - 2z = 0, \\ & 5x + 9y + 23z = 0 \end{aligned}$$

Solution:

Matrix form

$$AX = B, \text{ where}$$

$$A = \begin{bmatrix} 3 & 2 & 7 \\ 4 & -3 & -2 \\ 5 & 9 & 23 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} 3 & 2 & 7 & 0 \\ 4 & -3 & -2 & 0 \\ 5 & 9 & 23 & 0 \end{array} \right]$$

Apply row operations: $R_2 \rightarrow 3R_2 - 4R_1$, $R_3 \rightarrow 3R_3 - 5R_1$

$$\text{Then } \left[\begin{array}{ccc|c} 3 & 2 & 7 & 0 \\ 0 & -17 & -34 & 0 \\ 0 & 17 & 34 & 0 \end{array} \right]$$

Now $R_3 \rightarrow R_3 + R_2$

$$\text{So the echelon form is } \left[\begin{array}{ccc|c} 3 & 2 & 7 & 0 \\ 0 & -17 & -34 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Therefore, $\text{rank}(A) = \text{rank}([A|B]) = 2 < n$.

The system is consistent and has non-trivial solutions.

Writing the equivalent equations from echelon form:

$$3x + 2y + 7z = 0 \quad \dots (1)$$

$$-17y - 34z = 0 \quad \dots (2)$$

Put $z = t$.

$$\text{From (2): } -17y = 34t$$

$$\text{So } y = -2t.$$

$$\text{From (1): } 3x + 2(-2t) + 7t = 0$$

$$3x - 4t + 7t = 0$$

$$3x + 3t = 0$$

$$\text{So } x = -t.$$

Hence, $(x, y, z) = (-t, -2t, t)$, for all $t \in \mathbb{R}$.

Exercise 1.7

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Question 1

(ii) Solve the following system of homogeneous equations.

$$2x + 3y - z = 0,$$

$$x - y - 2z = 0,$$

$$3x + y + 3z = 0$$

Solution:

Matrix form

$$AX = B, \text{ where}$$

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & -1 & -2 \\ 3 & 1 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Augmented matrix

$$[A|B] = \left[\begin{array}{ccc|c} 2 & 3 & -1 & 0 \\ 1 & -1 & -2 & 0 \\ 3 & 1 & 3 & 0 \end{array} \right]$$

Apply row operations: $R_1 \leftrightarrow R_2$

$$\text{Then } \left[\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ 2 & 3 & -1 & 0 \\ 3 & 1 & 3 & 0 \end{array} \right]$$

Now $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$

$$\text{So we get } \left[\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ 0 & 5 & 3 & 0 \\ 0 & 4 & 9 & 0 \end{array} \right]$$

Next $R_3 \rightarrow 5R_3 - 4R_2$

$$\text{So the echelon form is } \left[\begin{array}{ccc|c} 1 & -1 & -2 & 0 \\ 0 & 5 & 3 & 0 \\ 0 & 0 & 33 & 0 \end{array} \right]$$

Therefore, $\text{rank}(A) = \text{rank}([A|B]) = 3 = n$.

The system is consistent and has only the trivial solution.

From the echelon form,

$$33z = 0, \text{ so } z = 0.$$

$$5y + 3z = 0, \text{ so } y = 0.$$

$$\text{From } x - y - 2z = 0, \text{ we get } x = 0.$$

$$\text{Hence, } x = 0, y = 0, z = 0.$$

Exercise 1.7

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Question 2

Determine the values of λ for which the following system of equations

$$x + y + 3z = 0,$$

$$4x + 3y + \lambda z = 0,$$

$$2x + y + 2z = 0$$

has

(i) a unique solution

(ii) a non-trivial solution.

Solution:

Since it is a homogeneous system, it is always consistent.
We test the rank of the coefficient matrix.

Matrix form

$$AX = B, \text{ where}$$

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 4 & 3 & \lambda \\ 2 & 1 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Augmented matrix

$$[A | B] = \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 4 & 3 & \lambda & 0 \\ 2 & 1 & 2 & 0 \end{array} \right]$$

Apply row operations: $R_2 \rightarrow R_2 - 4R_1$, $R_3 \rightarrow R_3 - 2R_1$

Then we get

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & \lambda - 12 & 0 \\ 0 & -1 & -4 & 0 \end{array} \right]$$

Now apply $R_3 \rightarrow R_3 - R_2$

So the echelon form becomes

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & \lambda - 12 & 0 \\ 0 & 0 & 8 - \lambda & 0 \end{array} \right]$$

Exercise 1.7

4/10

Question 2 — continued

From the echelon form

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & \lambda - 12 & 0 \\ 0 & 0 & 8 - \lambda & 0 \end{array} \right]$$

we discuss two cases.

Case (i): $\lambda \neq 8$

Then $8 - \lambda$ is not zero, so the third row is non-zero.

Hence, $\text{rank}(A) = \text{rank}([A|B]) = 3 = n$.

Therefore the system has a unique solution.

Because the system is homogeneous, the unique solution is the trivial solution:

$$x = 0, \quad y = 0, \quad z = 0.$$

Case (ii): $\lambda = 8$

Then the third row becomes zero, so

$$\text{rank}(A) = \text{rank}([A|B]) = 2 < 3.$$

Therefore the system has non-trivial solutions.

For $\lambda = 8$, the equations from echelon form are

$$x + y + 3z = 0 \quad \dots (1)$$

$$-y - 4z = 0 \quad \dots (2)$$

Put $z = t$.

From (2), $y = -4t$.

From (1), $x - 4t + 3t = 0$, so $x = t$.

Hence, the non-trivial solutions are

$$(x, y, z) = (t, -4t, t), \text{ for all } t \in \mathbb{R}.$$

Final answer:

Unique solution when $\lambda \neq 8$.

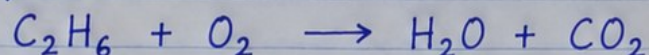
Non-trivial solutions when $\lambda = 8$.

Exercise 1.7

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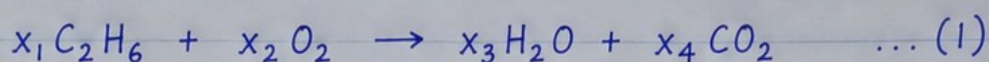
Question 3

By using Gaussian elimination method, balance the chemical reaction equation:



Solution:

Let the balanced form be



By balancing the atoms on both sides, we get:

For carbon:

$$2x_1 - x_4 = 0 \quad \dots (2)$$

For hydrogen:

$$6x_1 - 2x_3 = 0$$

Dividing by 2,

$$3x_1 - x_3 = 0 \quad \dots (3)$$

For oxygen:

$$2x_2 - x_3 - 2x_4 = 0 \quad \dots (4)$$

Thus, equations (2), (3), and (4) form a homogeneous system in the unknowns x_1, x_2, x_3, x_4 .

Augmented matrix

$$[A | B] = \left[\begin{array}{cccc|c} 2 & 0 & 0 & -1 & 0 \\ 3 & 0 & -1 & 0 & 0 \\ 0 & 2 & -1 & -2 & 0 \end{array} \right]$$

We apply Gaussian elimination on the next page.

Exercise 1.7

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Question 3 - continued

Start with the augmented matrix

$$\left[\begin{array}{cccc|c} 2 & 0 & 0 & -1 & 0 \\ 3 & 0 & -1 & 0 & 0 \\ 0 & 2 & -1 & -2 & 0 \end{array} \right]$$

Apply row operation $R_1 \leftrightarrow R_2$.

Then

$$\left[\begin{array}{cccc|c} 3 & 0 & -1 & 0 & 0 \\ 2 & 0 & 0 & -1 & 0 \\ 0 & 2 & -1 & -2 & 0 \end{array} \right]$$

Now apply $R_2 \rightarrow 3R_2 - 2R_1$.

So we get

$$\left[\begin{array}{cccc|c} 3 & 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & -3 & 0 \\ 0 & 2 & -1 & -2 & 0 \end{array} \right]$$

Next, interchange the last two rows to write the matrix in echelon order:

$$R_2 \leftrightarrow R_3.$$

Hence the echelon form is

$$\left[\begin{array}{cccc|c} 3 & 0 & -1 & 0 & 0 \\ 0 & 2 & -1 & -2 & 0 \\ 0 & 0 & 2 & -3 & 0 \end{array} \right]$$

Therefore, $\text{rank}(A) = \text{rank}([A|B]) = 3 < 4$.

So the system is consistent and has infinitely many solutions.

Continue the solution on the next page.

Exercise 1.7

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Question 3

From the echelon form, we get the equations

$$3x_1 - x_3 = 0 \quad \dots (1)$$

$$2x_2 - x_3 - 2x_4 = 0 \quad \dots (2)$$

$$2x_3 - 3x_4 = 0 \quad \dots (3)$$

Put $x_4 = t$.

From (3), $2x_3 = 3t$, so $x_3 = \frac{3}{2}t$.

From (1), $3x_1 = x_3 = \frac{3}{2}t$, so $x_1 = \frac{t}{2}$.

From (2),

$$2x_2 - \frac{3}{2}t - 2t = 0$$

$$2x_2 = \frac{3}{2}t + 2t = \frac{7}{2}t$$

$$x_2 = \frac{7}{4}t.$$

Hence,

$$(x_1, x_2, x_3, x_4) = \left(\frac{t}{2}, \frac{7t}{4}, \frac{3t}{2}, t \right), \text{ for all } t \in \mathbb{R}.$$

Since the coefficients in a chemical equation must be positive integers, choose $t = 4$.

Then,

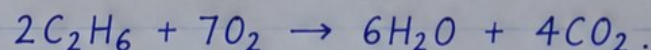
$$x_1 = 2,$$

$$x_2 = 7,$$

$$x_3 = 6,$$

$$x_4 = 4.$$

Therefore, the balanced chemical equation is



Exercise 1.7

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Summary of Question 1

Question 1 (i):

System:

$$3x + 2y + 7z = 0,$$

$$4x - 3y - 2z = 0,$$

$$5x + 9y + 23z = 0.$$

Result:

$$\text{rank}(A) = \text{rank}([A|B]) = 2 < 3.$$

So the system is consistent and has non-trivial solutions.

General solution:

$$(x, y, z) = (-t, -2t, t), \text{ for all } t \in \mathbb{R}.$$

Question 1 (ii):

System:

$$2x + 3y - z = 0,$$

$$x - y - 2z = 0,$$

$$3x + y + 3z = 0.$$

Result:

$$\text{rank}(A) = \text{rank}([A|B]) = 3 = n.$$

So the system is consistent and has only the trivial solution.

Final solution:

$$x = 0, \quad y = 0, \quad z = 0.$$

Exercise 1.7

9/10

Important notes and results

For a homogeneous system $AX = 0$:

1. If $\text{rank}(A) = n$, the system has only the trivial solution.
2. If $\text{rank}(A) < n$, the system has non-trivial solutions.

Application to Question 2:

For the system

$$\begin{aligned}x + y + 3z &= 0, \\4x + 3y + \lambda z &= 0, \\2x + y + 2z &= 0,\end{aligned}$$

we obtained the echelon form with the last entry $8 - \lambda$.

Therefore:

If $\lambda \neq 8$, then $\text{rank}(A) = 3$, so the unique solution is
 $x = y = z = 0$.

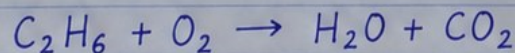
If $\lambda = 8$, then $\text{rank}(A) = 2 < 3$, so the system has non-trivial solutions.

One such general solution is

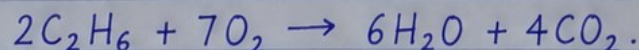
$$(x, y, z) = (t, -4t, t), \text{ for all } t \in \mathbb{R}.$$

Application to Question 3:

The balanced form of



is



Exercise 1.7

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Final Answer Sheet

Question 1 (i):

General non-trivial solution:

$$(x, y, z) = (-t, -2t, t), \text{ for all } t \in \mathbb{R}.$$

Question 1 (ii):

Trivial solution only:

$$x = 0, \quad y = 0, \quad z = 0.$$

Question 2:

$$\text{For } x + y + 3z = 0,$$

$$4x + 3y + \lambda z = 0,$$

$$2x + y + 2z = 0:$$

Unique solution when $\lambda \neq 8$.

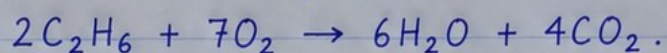
Non-trivial solutions when $\lambda = 8$.

For $\lambda = 8$, one general solution is

$$(x, y, z) = (t, -4t, t).$$

Question 3:

Balanced chemical equation:



Completed in 10 handwritten pages.

Exercise 1.8

1/10

Choose the Most Suitable Answer

Question 1

If $|\text{adj}(\text{adj } A)| = |A|^9$, then the order of the square matrix A is:

- (a) 3 (b) 4 (c) 2 (d) 5

Solution: (b) 4

Hint:

$$\begin{aligned}\text{Since } |\text{adj}(\text{adj } A)| &= |A|^{(n-1)^2} \\ (n-1)^2 &= 9 = 3^2 \\ n-1 &= 3 \\ n &= 4\end{aligned}$$

Question 2

If A is a 3×3 non-singular matrix such that $AA^T = A^T A$ and $B = A^{-1}A^T$, then $BB^T =$

- (a) A (b) B (c) I_3 (d) B^T

Solution: (c) I_3

Hint:

$$\begin{aligned}B^T &= (A^{-1}A^T)^T = A(A^{-1})^T \\ BB^T &= (A^{-1}A^T) \cdot A(A^{-1})^T \\ &= A^{-1}(A^T A)(A^{-1})^T \\ &= A^{-1}(AA^T)(A^{-1})^T \\ &= I_3\end{aligned}$$

Question 3

If $A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$, $B = \text{adj } A$ and $C = 3A$, then $\frac{|\text{adj } B|}{|C|} =$

- (a) $\frac{1}{3}$ (b) $\frac{1}{9}$ (c) $\frac{1}{4}$ (d) 1

Solution: (b) $\frac{1}{9}$

Hint:

$$\begin{aligned}\frac{|\text{adj } B|}{|C|} &= \frac{|\text{adj}(\text{adj } A)|}{|3A|} \\ &= \frac{|A|^{(n-1)^2}}{3^n |A|}\end{aligned}$$

For $n=2$, this becomes $\frac{|A|}{3^2 |A|} = \frac{1}{9}$

Exercise 1.8

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Question 4

If $A \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$, then A is

- (a) $\begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix}$

Solution: (c) $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$

Hint:

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Then $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$

So $a+b=6, \quad -2a+4b=0,$
 $c+d=0, \quad -2c+4d=6.$

Solving, $a=4, b=2, c=-1, d=1.$

Hence $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}.$

Question 5

If $A = \begin{bmatrix} 7 & 3 \\ 4 & 4 \end{bmatrix}$, then $9I_2 - A =$

- (a) A^{-1} (b) $\frac{1}{2}A^{-1}$ (c) $3A^{-1}$ (d) $2A^{-1}$

Solution: (d) $2A^{-1}$

Hint:

$|A| = 14 - 12 = 2.$

$\text{adj } A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}, \quad A^{-1} = \frac{1}{2} \text{adj } A.$

So $2A^{-1} = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}.$

Also $9I_2 - A = \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}.$

Therefore $9I_2 - A = 2A^{-1}.$

Question 6

If $A = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$, then $|\text{adj}(AB)| =$

- (a) -40 (b) -80 (c) -60 (d) -20

Solution: (b) -80

Hint: $AB = \begin{bmatrix} 2 & 8 \\ 11 & 4 \end{bmatrix}.$ For a 2×2 matrix M , $|\text{adj } M| = |M|.$

So $|\text{adj}(AB)| = |AB| = (2)(4) - (8)(11) = 8 - 88 = -80.$

Exercise 1.8

3/10

Question 7

If $P = \begin{bmatrix} 1 & 1 & 2 \\ x & 3 & 4 \\ 0 & 0 & -2 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then x is

- (a) 15 (b) 12 (c) 14 (d) 11

Solution: (d) 11

Hint:

Since $P = \text{adj} A$ and A is of order 3,

$$|\text{adj} A| = |A|^{n-1} = |A|^2 = 4^2 = 16.$$

$$\begin{aligned} \text{Now } |P| &= 1[3(-2) - 4(0)] - 1[x(-2) - 4(0)] + 2[x(0) - 3(0)] \\ &= -6 + 2x. \end{aligned}$$

$$\text{So } -6 + 2x = 16.$$

$$2x = 22.$$

$$\text{Therefore } x = 11.$$

Question 8

If $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -2 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ and $A^{-1} = [a_{ij}]$, then the value of a_{23} is

- (a) 0 (b) -2 (c) -3 (d) -1

Solution: (d) -1

Hint:

$$a_{23} = \frac{A_{32}}{|A|}.$$

$$\text{The cofactor } A_{32} = -\det \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix} = -(0+2) = -2.$$

$$\text{Also } |A| = 3(2) - 1(-2) - 1(4+2) = 6+2-6 = 2.$$

$$\text{Hence } a_{23} = \frac{-2}{2} = -1.$$

Question 9

If A , B and C are invertible matrices of some order, then which one of the following is not true?

- (a) $\text{adj} A = |A|A^{-1}$ (b) $\text{adj}(AB) = (\text{adj} A)(\text{adj} B)$
(c) $\det A^{-1} = (\det A)^{-1}$ (d) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

Solution: (b) $\text{adj}(AB) = (\text{adj} A)(\text{adj} B)$

Hint:

The correct identity is $\text{adj}(AB) = (\text{adj} B)(\text{adj} A)$, not $(\text{adj} A)(\text{adj} B)$.

Exercise 1.8

4/10

Question 10

If $(AB)^{-1} = \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$, then $B^{-1} =$

- (a) $\begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 8 & 5 \\ 3 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 8 & -5 \\ -3 & 2 \end{bmatrix}$

Solution: (a) $\begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix}$

Hint:

$$(AB)^{-1} = B^{-1}A^{-1}.$$

$$\text{So } B^{-1} = (AB)^{-1}A.$$

$$\text{Now } A = (A^{-1})^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}.$$

Hence

$$\begin{aligned} B^{-1} &= \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 36 - 34 & 12 - 17 \\ -57 + 54 & -19 + 27 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix}. \end{aligned}$$

Question 11

If $A^T A^{-1}$ is symmetric, then $A^2 =$

- (a) A^{-1} (b) $(A^T)^2$ (c) A^T (d) $(A^{-1})^2$

Solution: (b) $(A^T)^2$

Hint:

Since $A^T A^{-1}$ is symmetric,

$$(A^T A^{-1})^T = A^T A^{-1}.$$

$$\text{So } (A^{-1})^T A = A^T A^{-1}.$$

Pre-multiplying by A^T and post-multiplying by A ,
we get $A^2 = (A^T)^2$.

Question 12

If A is a non-singular matrix such that $A^{-1} = \begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$, then $(A^T)^{-1} =$

- (a) $\begin{bmatrix} -5 & 3 \\ 2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} -1 & -3 \\ 2 & 5 \end{bmatrix}$ (d) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$

Solution: $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$

Hint:

$$(A^T)^{-1} = (A^{-1})^T.$$

$$\text{Therefore } (A^T)^{-1} = \begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}.$$

Exercise 1.8

5/10

Question 13

If $A = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ x & \frac{3}{5} \end{bmatrix}$ and $A^T = A^{-1}$, then the value of x is

- (a) $-4/5$ (b) $-3/5$ (c) $3/5$ (d) $4/5$

Solution: (a) $-4/5$

Hint:

Since $A^T = A^{-1}$, we have $AA^T = I$.

Using the off-diagonal entry,

$$(3/5)x + (4/5)(3/5) = 0.$$

$$\text{So } (3/5)x + 12/25 = 0.$$

$$(3/5)x = -12/25.$$

$$\text{Hence } x = -4/5.$$

Question 14

Answer the following objective question:

- (a) $\frac{\cos^2 \theta}{2} A$ (b) $\frac{\cos^2 \theta}{2} A^T$ (c) $\cos^2 \theta I$ (d) $\frac{\sin^2 \theta}{2} A$

Solution: (b) $\frac{\cos^2 \theta}{2} A^T$

Question 15

Answer the following objective question:

- (a) 0 (b) $\sin \theta$ (c) $\cos \theta$ (d) 1

Solution: (d) 1

Hint:

Let the required value be k .

Using the identity $\cos^2 \theta + \sin^2 \theta = k$,

we get $k = 1$.

Exercise 1.8

6/10

Question 16

If $A = \begin{bmatrix} 2 & 5 \\ 3 & -2 \end{bmatrix}$ is such that $\lambda A^{-1} = A$, then λ is

- (a) 17 (b) 14 (c) 19 (d) 21

Solution: (c) 19

Hint:

From $\lambda A^{-1} = A$, multiplying by A gives $\lambda I = A^2$.

Now

$$\begin{aligned} A^2 &= \begin{bmatrix} 2 & 5 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 3 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 19 & 0 \\ 0 & 19 \end{bmatrix} = 19I. \end{aligned}$$

Hence $\lambda = 19$.

Question 17

If $\text{adj } A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$ and $\text{adj } B = \begin{bmatrix} 1 & -3 \\ -2 & 1 \end{bmatrix}$, then $\text{adj}(AB)$ is:

- (a) (b) (c) (d)

Solution: (b)

Hint:

$$\text{adj}(AB) = (\text{adj } B)(\text{adj } A).$$

So multiply the two given matrices and choose the matching option.

Question 18

(Rank question - answer only)

- (a) 1 (b) 2 (c) 4 (d) 4

Solution: (a) 1

Hint:

Number of non-zero rows = 1.

Therefore $\text{rank}(A) = 1$.

Exercise 1.8

7/10

Question 19

If $x^a y^b = e^m$ and $x^c y^d = e^n$, then x and y are

- (a) $e^{\Delta_2/\Delta_1}, e^{\Delta_3/\Delta_1}$ (b) $\log(\Delta_1/\Delta_3), \log(\Delta_3/\Delta_3)$
(c) $\log(\Delta_2/\Delta_1), \log(\Delta_3/\Delta_1)$ (d) $e^{\Delta_1/\Delta_3}, e^{\Delta_2/\Delta_3}$

Solution: (d) $e^{\Delta_1/\Delta_3}, e^{\Delta_2/\Delta_3}$

Hint:

Taking logarithms,

$$a \log x + b \log y = m,$$

$$c \log x + d \log y = n.$$

Solving this linear system by Cramer's rule gives

$$\log x = \Delta_1/\Delta_3 \quad \text{and} \quad \log y = \Delta_2/\Delta_3.$$

$$\text{Hence } x = e^{\Delta_1/\Delta_3}, y = e^{\Delta_2/\Delta_3}.$$

Question 20

Which of the following is/are correct?

- (i) Adjoint of a symmetric matrix is also a symmetric matrix.
(ii) Adjoint of a diagonal matrix is also a diagonal matrix.
(iii) If A is a square matrix of order n and λ is a scalar, then $\text{adj}(\lambda A) = \lambda^n \text{adj}(A)$.
(iv) $A(\text{adj } A) = (\text{adj } A)A = |A|I$.
(a) Only (i) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i), (ii) and (iv)

Solution: (d) (i), (ii) and (iv)

Hint:

Statement (iii) is false; the correct formula is $\text{adj}(\lambda A) = \lambda^{n-1} \text{adj}(A)$.

Question 21

If $\text{rank}(A) = \text{rank}([A|B])$, then the system $AX = B$ is

- (a) consistent and has a unique solution (b) consistent
(c) consistent and has infinitely many solutions (d) inconsistent

Solution: (b) consistent

Hint:

Equality of $\text{rank}(A)$ and $\text{rank}([A|B])$ is the condition for consistency.

Exercise 1.8

8/10

Question 22

If $0 \leq \theta \leq \pi$ and the system of equations

$$x + (\sin \theta)y - (\cos \theta)z = 0,$$

$$(\cos \theta)x - y + z = 0,$$

$$(\sin \theta)x + y + z = 0$$

has a non-trivial solution, then θ is

- (a) $2\pi/3$ (b) $3\pi/4$ (c) $5\pi/6$ (d) $\pi/4$

Solution: (d) $\pi/4$

Hint:

A homogeneous system $AX = 0$ has a non-trivial solution when $|A| = 0$.

So,

$$\begin{vmatrix} 1 & \sin \theta & -\cos \theta \\ \cos \theta & -1 & 1 \\ \sin \theta & 1 & 1 \end{vmatrix} = 0.$$

This simplifies to $\sin^2 \theta - \cos^2 \theta = 0$.

Hence $\sin \theta = \cos \theta$, so $\theta = \pi/4$.

Question 23

Answer the following parameter question:

(a) $\lambda = 7, \mu \neq -5$ (b) $\lambda = -7, \mu = 5$

(c) $\lambda \neq 7, \mu \neq -5$ (d) $\lambda = 7, \mu = -5$

Solution: (d) $\lambda = 7, \mu = -5$

Hint:

$$\text{rank}(A) = \text{rank}([A|B]) = 2.$$

$$\text{So } \lambda - 7 = 0 \text{ and } \mu + 5 = 0.$$

$$\text{Therefore } \lambda = 7 \text{ and } \mu = -5.$$

Exercise 1.8

9/10

Question 24

Answer the following objective question:

- (a) 2 (b) 4 (c) 3 (d) 1

Solution: (b) 4

Hint:

$$B = A^{-1}.$$

So the required value is 4.

Question 25

Find $\text{adj}(\text{adj } A)$.

Solution: (a) $\begin{bmatrix} 3 & 2 & 0 \\ -3 & -3 & -1 \\ 4 & 4 & 1 \end{bmatrix}$

Hint:

Use the formula $\text{adj}(\text{adj } A) = |A|^{n-2} A$.

From the given matrix in the exercise, $|A| = 1$.

So $\text{adj}(\text{adj } A) = A$.

Hence the answer is $\begin{bmatrix} 3 & 2 & 0 \\ -3 & -3 & -1 \\ 4 & 4 & 1 \end{bmatrix}$.

All 25 questions are now covered. Final answer sheet on the next page.

Exercise 1.8

10/10

Final Answer Sheet

- Q1 → (b) 4
Q2 → (c) I_3
Q3 → (b) $\frac{1}{9}$
Q4 → (c) $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$
Q5 → (d) $2A^{-1}$
Q6 → (b) -80
Q7 → (d) 11
Q8 → (d) -1
Q9 → (b)
Q10 → (a) $\begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix}$
Q11 → (b) $(A^T)^2$
Q12 → $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$
Q13 → (a) $-\frac{4}{5}$
Q14 → (b) $\frac{\cos^2 \theta}{2} A^T$
Q15 → (d) 1
Q16 → (c) 19
Q17 → (b)
Q18 → (a) 1
Q19 → (d) $(e^{\Delta_1/\Delta_3}, e^{\Delta_2/\Delta_3})$
Q20 → (d) (i), (ii) and (iv)
Q21 → (b) consistent
Q22 → (d) $\frac{\pi}{4}$
Q23 → (d) $\lambda = 7, \mu = -5$
Q24 → (b) 4
Q25 → (a) $\begin{bmatrix} 3 & 2 & 0 \\ -3 & -3 & -1 \\ 4 & 4 & 1 \end{bmatrix}$

Completed in 10 handwritten pages.